

COURSE: CS 301 — Algorithms

Credits: 4 | Department: Computer Science

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PREREQUISITES: CS 201 (minimum grade C) AND MATH 231

CO-REQUISITES: None

OFFERED: Fall, Spring

DESCRIPTION:

Design and analysis of algorithms. Topics include divide-and-conquer, dynamic programming, greedy algorithms, network flow, randomized algorithms, approximation algorithms, and advanced graph algorithms. Strong emphasis on mathematical analysis and proof of correctness. Students write weekly problem sets and two programming assignments.

MINIMUM GRADE TO SATISFY PREREQUISITE FOR CS 351 (ML Foundations): C

GRADING: Problem sets 45%, Programming assignments 25%, Midterm 15%, Final 15%.

NOTE: This is a mathematically rigorous course. Students who have not completed MATH 231 with at least a C should reconsider enrollment.